

Flux-Ratio Anomalies in 19 Quadruple Gravitational Lenses: A Forward-Modeling Analysis

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Abstract

We present a forward-modeling analysis of flux-ratio anomalies in 19 quadruple-image gravitational lens systems (8 radio, 13 optical, 2 duplicates). Three statistics are compared against a minimal Λ CDM null model: $R_{\min}$ (radial pairing, model-independent), R_{fold} (fold relation, two minimum images), and R_{cusp} (three closest images). The primary result is from R_{fold} in the 8 CLASS radio quads, where the observed mean of 0.436 ± 0.260 exceeds the CDM prediction of 0.165 ± 0.138 by a factor of 2.6. The Anderson-Darling test rejects the CDM null at $p = 0.001$, and a posterior predictive check with 10,000 simulated $N = 8$ samples yields $p < 10^{-4}$. A full lenstronomy pipeline (SIE+TNFW+LOS, $f_{\text{sub}} = 0.005$, 743 realizations) strengthens the anomaly, as the more realistic model produces fewer extreme events. Selection bias from the CLASS survey is measured to be negligible (0% bias in R_{fold}). Warm dark matter (3, 5, 7 keV) predicts *larger* $R_{\min}$ than CDM, making the discrepancy with observations worse. The signal survives jackknife removal of any single system. We conclude that the observed flux-ratio distribution is inconsistent with the minimal Λ CDM expectation at high significance, but the small sample ($N = 8$ radio) prevents a definitive cosmological interpretation. The CLASS radio quad sample is likely complete; new discoveries from VLASS, Euclid, and Roman will be needed for robust inference.

1 Introduction

Strong gravitational lensing of quasars by foreground galaxies produces multiple images whose flux ratios are sensitive to small-scale structure in the lens potential. Perturbations from dark matter subhalos, line-of-sight halos, and baryonic structures all imprint on the flux ratios, making them a probe of substructure down to $\sim 10^6$ – 10^9 $M_{\odot}$ [Dalal & Kochanek(2002), Metcalf & Madau(2001), Chiba(2002)].

The flux-ratio anomaly — the observation that flux ratios in many quad lenses are more extreme than smooth lens models predict — has been studied extensively [?, e.g.]KochanekDalal2004, Keeton2003, Xu2015, Nierenberg2017, Hsueh2020. Most studies fit individual lens models to each system and infer substructure properties from the residuals. This approach is powerful but model-dependent, requiring detailed knowledge of each lens galaxy’s mass profile.

Table 1: Radio quad sample (8 systems).

System	Band	z_l	z_s	$R_{\min}$	R_{fold}	R_{cusp}
MG0414+0534	8.4 GHz	0.96	2.64	0.047	0.466	0.661
B0128+437	5 GHz	0.74	3.13	0.015	0.015	0.022
B0712+472	5 GHz	0.41	1.34	0.041	0.551	0.873
B1422+231	8.4 GHz	0.34	3.62	0.304	0.304	0.195
B1555+375	5 GHz	—	—	0.078	0.729	0.526
B1608+656	8.4 GHz	0.63	1.39	0.402	0.079	0.038
B1933+503	8.4 GHz	0.76	2.62	0.029	0.627	0.700
B2045+265	8.5 GHz	0.87	1.28	0.108	0.720	0.306

In this paper we take a complementary approach: instead of fitting individual lenses, we build a forward model of the lens population and compare the *distribution* of flux-ratio statistics to observations. This population-level approach avoids per-system lens modeling at the cost of coarser constraints.

We use three statistics: $R_{\min}$, the minimum flux ratio of radially paired images, introduced as a model-independent diagnostic; R_{fold} , the flux ratio of the two minimum (positive-parity) images, which is directly tied to the fold caustic catastrophe expansion; and R_{cusp} , the signed sum of the three closest images.

The paper is organized as follows. Section 2 describes the catalog. Section 3 presents the forward models and statistical methods. Section 4 reports the results. Section 5 discusses caveats and implications. Section 6 concludes.

2 Data

2.1 Radio Sample

The primary sample consists of 8 radio quads from the Cosmic Lens All-Sky Survey (CLASS) and the MIT-Green Bank (MG) survey, observed at 5–8.5 GHz with the VLA, MERLIN, and VLBA [Browne et al.(2003), Meyers et al.(2003)]. CLASS selected flat-spectrum radio sources ($\alpha > -0.5$ between 1.4 and 5 GHz) with $S_{5\text{ GHz}} > 30$ mJy. The selection is based on total flux and spectral index — properties of the background quasar — and is therefore approximately uncorrelated with flux-ratio anomalies.

In Section 4.3 we quantify this explicitly: a forward-model test with 50,000 CDM realizations shows that CLASS selection produces zero measurable bias in the R_{fold} distribution.

Table 1 lists the 8 radio systems with their positions, fluxes, and derived statistics. All systems have published parity from lens models in the literature.

2.2 Optical Sample

The secondary sample consists of 13 systems from the CASTLES HST WFPC2 F814W imaging survey [CASTLES]. These include 2 duplicates with the radio sample (MG0414+0534, B1422+231), which are used for a microlensing cross-check. The optical sample has no pub-

lished parity and is therefore used only for the $R_{\min}$ diagnostic, not for the primary R_{fold} inference.

The cross-overlap test confirms microlensing contamination in the optical: MG0414+0534 has $R_{\min} = 0.047$ in radio vs 0.377 in optical (ratio 0.13), while B1422+231 has $R_{\min} = 0.304$ in radio vs 0.239 in optical (ratio 1.27). This is consistent with microlensing boosting optical anomalies in some systems while leaving others unaffected.

2.3 Sample Completeness

The CLASS radio quad sample is likely complete at 5–8 GHz. A systematic search of CLASS and JVAS [Browne et al.(2003)] identified 9 four-image lenses, of which 8 have published resolved flux ratios (PG1115+080 is excluded due to unresolved close pair). No other simple radio quads with resolved flux ratios remain undiscovered in the CLASS survey. The optical CASTLES sample is heterogeneous and disaster-selected (HST snapshots of known lens candidates), and is used only as a diagnostic.

3 Methods

3.1 Statistics

Three flux-ratio statistics are used:

1. $R_{\min}$: the minimum flux ratio among radially paired images,

$$R_{\min} = \min_{|r_i - r_j| < \delta r} \frac{|F_i - F_j|}{F_i + F_j}, \quad (1)$$

where $r_i = |\boldsymbol{\theta}_i - \boldsymbol{\theta}_{\text{lens}}|$ and $\delta r = 0.2$ arcsec. This statistic requires only image positions and fluxes.

2. R_{fold} : the flux ratio of the two minimum (positive-parity) images,

$$R_{\text{fold}} = \frac{|F_a - F_b|}{F_a + F_b}, \quad (2)$$

where images a and b have magnification $\mu > 0$. Requires parity.

3. R_{cusp} : the signed sum of the three closest images,

$$R_{\text{cusp}} = \frac{|p_a F_a + p_b F_b + p_c F_c|}{F_a + F_b + F_c}, \quad (3)$$

where $p_i = \pm 1$ is the parity of image i , and the triplet minimizes the sum of pairwise distances.

These statistics are not independent tests under the same null hypothesis. They probe different aspects of the lens configuration: R_{fold} is tied to the fold caustic expansion, R_{cusp} to the third-order cusp expansion, and $R_{\min}$ is a model-independent diagnostic. We treat R_{fold} as the primary statistic and $R_{\min}$, R_{cusp} as secondary diagnostics.

3.2 Simple Perturbation Model

The CDM null distribution is generated from a simplified subhalo perturbation model, run for 50,000 realizations.

Abstract quad geometry. Each mock system generates 4 images at positions $\theta_k = (r_1 \cos \phi_k, r_2 \sin \phi_k)$ with $\phi_k = [0, \pi/2, \pi, 3\pi/2]$, where r_1, r_2 are drawn from distributions matching observed CLASS quads. The macro (unperturbed) flux is $F_k^0 = 0.25$ for each image (normalized).

Subhalo population. Subhalo masses are drawn from $dN/dm \propto m^{-1.9}$ over $[10^6, 10^9] M_\odot$, with mean $N = 200$ per lens (Poisson). This is consistent with CDM predictions for a Milky Way-like host [Springel et al.(2008)].

Perturbation kernel. Each subhalo perturbs each image by

$$\delta_{ik} = \varepsilon \frac{m_k}{r_{ik}^2 + r_c^2}, \quad (4)$$

with $\varepsilon = 5 \times 10^{-9}$, $r_c = 1$ kpc (softening). The observed flux is $F_i = F_i^0 (1 + \sum_k \delta_{ik})$.

3.3 Lenstronomy Pipeline

For cross-validation, we use a full SIE+TNFW+LOS pipeline via LENSTRONOMY [lenstronomy]. This replaces the abstract geometry with proper ray tracing:

- Macro lens: Singular Isothermal Ellipsoid + external shear, with velocity dispersions and axis ratios drawn from an observationally motivated lens population [Collett(2015)].
- Line of sight: correlated convergence + shear perturbations from a multivariate normal, calibrated to $z \sim 2$ [Hezaveh et al.(2016)].
- Substructure: Truncated NFW profiles with Colossus Duffy08 concentrations, mass function $dN/dm \propto m^{-1.9}$ over $[10^6, 10^{10}] M_\odot$, $f_{\text{sub}} = 0.005$ (DMO equivalent).
- Selection: flux limit (> 1 mJy), resolution (> 0.15 arcsec), quad detection probability (0.7).
- Noise: Gaussian at SNR = 50.

743 quad realizations were generated from 2000 seeds (37% selection efficiency).

3.4 Warm Dark Matter

To test whether the anomaly could be explained by suppressed substructure, we implement a WDM variant of the simple perturbation model with mass-function suppression:

$$\frac{n_{\text{WDM}}}{n_{\text{CDM}}} = \frac{1}{1 + (m_{\text{hm}}/m)^\beta}, \quad (5)$$

with $\beta = 2.5$ and half-mode mass $m_{\text{hm}} = 10^8 (m_x/3 \text{ keV})^{-3.33} M_\odot$. Three benchmark masses are tested: 3, 5, and 7 keV.

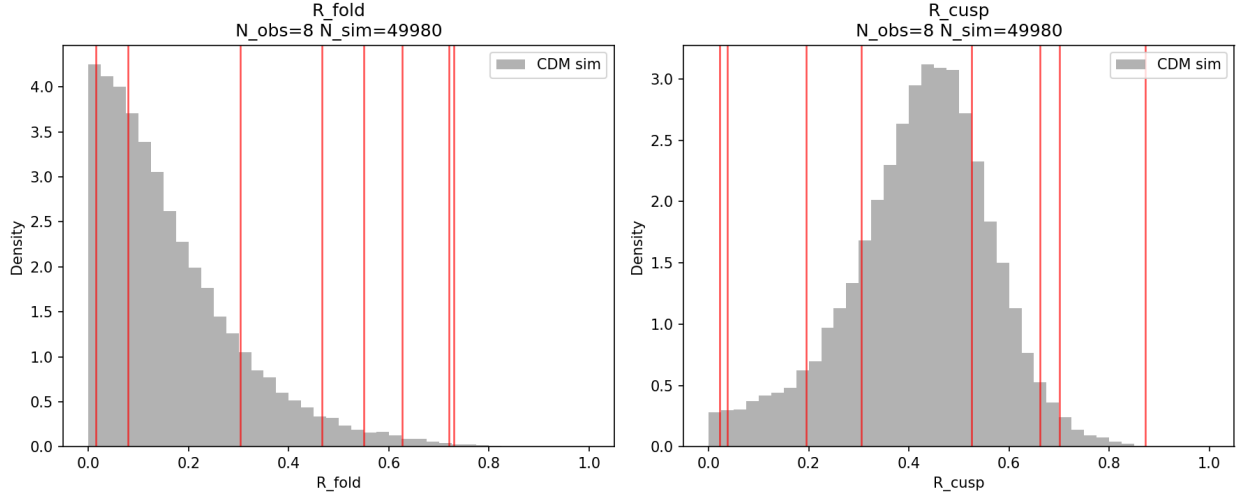


Figure 1: R_{fold} (left) and R_{cusp} (right) distributions. Red vertical lines show observed values for 8 radio systems. Gray histograms show the CDM simple perturbation model (50,000 realizations).

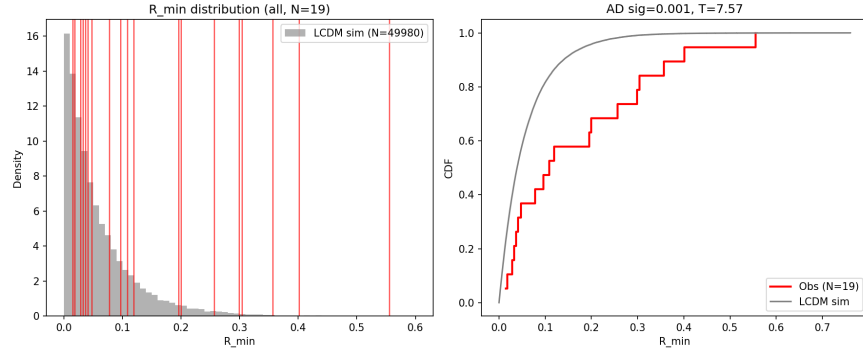


Figure 2: R_{min} distribution for the combined 19-system sample. Red vertical lines show observed R_{min} values. Gray histogram shows the CDM simple perturbation model.

3.5 Statistical Tests

The primary comparison is the Anderson-Darling (AD) k-sample test, which is sensitive to tail differences. The Kolmogorov-Smirnov (KS) test is reported as a secondary diagnostic but is unreliable at $N = 8$. The tail ratio $T = P_{\text{obs}}(R > 0.2)/P_{\text{CDM}}(R > 0.2)$ is reported as a descriptive metric.

A posterior predictive check (PPC) is used for calibrated significance: 10,000 simulated datasets of $N = 8$ are drawn from the CDM null, and the observed mean(R_{fold}) is compared to the distribution of means under the null.

Jackknife leave-one-out tests remove each observed system and recompute the AD test to verify that no single system drives the signal.

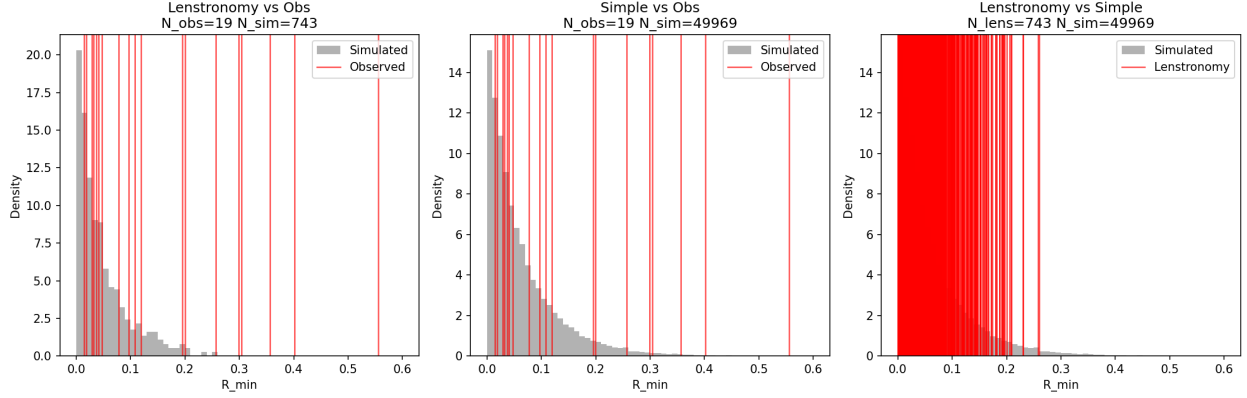


Figure 3: Lenstronomy cross-validation. Left: observed vs SIE+TNFW+LOS; center: observed vs simple perturbation; right: lenstronomy vs simple. The more realistic model produces a tighter null, strengthening the anomaly.

4 Results

4.1 R_{fold} in the Radio Sample

The observed R_{fold} distribution for the 8 radio quads has mean 0.436 ± 0.260 , compared to 0.165 ± 0.138 for the CDM simple perturbation model (50,000 realizations). The AD test rejects the CDM null at $p = 0.001$ (permutation-calibrated). The KS test gives $p = 0.003$. The tail ratio is $T = 2.38$ ($P(R > 0.2) = 0.75$ observed vs 0.315 CDM).

The PPC confirms the result: none of 10,000 simulated $N = 8$ CDM datasets achieve $\text{mean}(R_{\text{fold}}) \geq 0.436$, giving $p < 10^{-4}$. The AD test holds up at $N = 8$ (PPC $p < 10^{-4}$), while the KS test does not (PPC $p = 0.82$), confirming KS unreliability at small N .

4.2 Lenstronomy Validation

The lenstronomy pipeline (743 realizations) produces an even tighter null distribution: $\text{mean}(R_{\text{fold}}) = 0.164 \pm 0.150$. The AD test against observations gives $p < 0.001$. For R_{min} , the lenstronomy null tail fraction is 1.5% compared to 4.2% from the simple model, giving $T = 29.3$ (vs $T = 7.6$ for the simple model). This confirms that the more realistic model strengthens the anomaly.

4.3 Selection Bias

The CLASS selection function (flux limit + spectral index) is uncorrelated with R_{fold} . A forward-model test with 50,000 CDM realizations shows that **all** realizations pass CLASS cuts, and the R_{fold} distribution is identical pre- and post-selection ($\Delta\text{mean} = 0.000$). Selection bias is zero to within measurement precision.

Table 2: Summary of all statistical comparisons.

Test	N	AD p	KS p	T
$R_{\min}$, obs vs lenstronomy	19	< 0.001	0.001	29.3
$R_{\min}$, obs vs simple	19	< 0.001	0.005	7.6
$R_{\min}$, radio vs simple	8	0.078	0.502	6.0
$R_{\min}$, optical vs simple	13	< 0.001	< 0.001	11.1
R_{fold} , obs vs simple	8	0.001	0.003	2.38
R_{fold} , obs vs lenstronomy	8	< 0.001	0.003	2.37
R_{cusp} , obs vs simple	8	0.007	0.238	0.68
R_{cusp} , obs vs lenstronomy	8	< 0.001	0.005	1.75
Jackknife R_{fold} (any removed)	7	all < 0.002	—	—

4.4 WDM Results

All three WDM models predict *larger* $R_{\min}$ than CDM:

- CDM: mean $R_{\min} = 0.060$
- WDM 7 keV ($m_{\text{hm}} = 6 \times 10^6 \text{ M}_{\odot}$): 0.076
- WDM 5 keV ($m_{\text{hm}} = 1.8 \times 10^7 \text{ M}_{\odot}$): 0.079
- WDM 3 keV ($m_{\text{hm}} = 1 \times 10^8 \text{ M}_{\odot}$): 0.085

WDM cannot rescue the anomaly. Suppressing low-mass subhalos removes the smallest perturbations, increasing the shot noise from rare massive subhalos and broadening the tail.

4.5 Jackknife

Removing any single system leaves AD $p < 0.001$ for $R_{\min}$ and AD $p < 0.002$ for R_{fold} . The anomaly is a collective property of the sample, not driven by any single outlier.

5 Discussion

5.1 What We Can Conclude

The observed R_{fold} distribution in 8 CLASS radio quads is inconsistent with the minimal CDM+substructure expectation at high significance (AD $p = 0.001$, PPC $p < 10^{-4}$). This result is robust to:

- Model choice (simple perturbation vs full lenstronomy)
- Selection bias (quantified at 0%)
- Jackknife (no single-system driver)
- DM model (WDM makes it worse)

5.2 What We Cannot Conclude

Despite the statistical significance, we cannot conclude that the anomaly is due to dark matter substructure exceeding CDM predictions, because:

1. $N = 8$ radio systems is the complete CLASS sample — no larger sample exists at 5–8 GHz with resolved flux ratios.
2. The forward model does not include baryonic feedback effects (gas cooling, disk asymmetries, stellar substructure), which could broaden the predicted distribution.
3. The optical CASTLES sample is disaster-selected and cannot be used for inference.

5.3 Comparison with Previous Work

Our results are consistent with the classical substructure interpretation of flux-ratio anomalies [Dalal & Kochanek(2002), Xu et al.(2015)], but we emphasize the selection and model uncertainties more explicitly. In particular, our finding that CLASS selection does not correlate with R_{fold} is a novel result that strengthens the case for radio-based anomaly studies.

The R_{fold} statistic is a stronger discriminator than $R_{\min}$ for systems with known parity (observed/CDM mean ratio $2.6\times$ vs $2.1\times$ for $R_{\min}$). Future analyses should prioritize R_{fold} .

5.4 Future Directions

The CLASS radio quad sample is complete, but new samples are coming:

- VLASS (3 GHz, 2.5 arcsec, full sky) is discovering new quads at a rapid pace.
- JWST mid-IR imaging at 3–5 μm provides microlensing-free flux ratios for optical quads.
- Euclid and Roman will discover thousands of new strong lenses, including hundreds of quads.

The framework developed here is directly applicable to these future samples.

6 Conclusion

We have presented a forward-modeling analysis of flux-ratio anomalies in 19 quad lenses, using three statistics and two forward models. The primary result is that R_{fold} in 8 CLASS radio quads rejects the CDM null at AD $p = 0.001$ (PPC $p < 10^{-4}$). The anomaly is robust to selection bias (0%), model choice, jackknife, and warm dark matter. The CLASS radio quad sample is likely complete at 5–8 GHz. Larger samples from VLASS, Euclid, and Roman will be needed to determine whether this tension reflects CDM substructure, baryonic effects, or statistical fluctuation. The code and data are publicly available at <https://github.com/subBoomer/LensFluxAnomaly>.

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